

23MCT406
ENVIRONMENTAL SCIENCE

Recycling and waste-to energy technologies

Main pathways for conversion of organic waste material to energy

1. Thermochemical conversion
2. Biochemical conversion
3. Physicochemical conversion
4. Anaerobic digestion
5. Biofuel production

1. Thermochemical conversion

- Thermochemical conversion includes ~ **incineration, pyrolysis and gasification**
- a) **Incineration** technology is the **controlled combustion** of waste
 - **Steam** generated in turn produces power through **steam turbines**
- b) **Pyrolysis and gasification** represent refined thermal treatment methods
 - **Pyrolysis** occur in **absence of air**
 - Thermal decomposition takes place in the presence of a limited amount of oxygen or air
 - Gas which is generated can then be used in either boilers or cleaned up and used in combustion **turbine/generators**

2. Biochemical conversion

- **Biomass** converted into gaseous or liquid fuels, such as **biogas** or **bioethanol**
- Conversion of biomass involves use of **bacteria, microorganisms and enzymes**
- Commonly used technologies ~ **Anaerobic digestion (or bio-methanation) and fermentation**
- In anaerobic digestion organic material is decomposed through the metabolic pathways of naturally occurring microorganisms in an oxygen depleted environment

3. Physicochemical conversion

- Done to improve physical and chemical properties of solid waste
- Municipal Solid Waste (MSW) is a poor-quality fuel
- To improve its calorific value of MSW pre-processing is needed
- Waste is converted into high-energy fuel pellets, known as **Refuse Derived Fuel (RDF)** which may be used in steam generation



4. Anaerobic digestion

- Stabilize organic waste in the **absence of air**
- Reliable technology for the treatment of wet, organic waste
- In an oxygen-free condition the production of **biogas** occurs
- Biogas can be used to produce both electricity and heat
- Organic material include ~ municipal solid waste, animal manure, poultry litter, food wastes, sewage and industrial wastes
- Used in gas supply networks

Anaerobic digestion occurs in four steps

- a) Hydrolysis: Large polymers are broken down by enzymes
- b) Fermentation: Acidogenic fermentations produce volatile fatty acids along with carbon dioxide and hydrogen
- c) Acetogenesis: Breakdown of volatile acids to acetate and hydrogen
- d) Methanogenesis: Acetate, formaldehyde, hydrogen and carbon dioxide are converted to methane and water

5. Biofuel production

- Fuel produced from waste resources including liquid fuels, such as ethanol, methanol, biodiesel and gaseous fuels such as hydrogen and methane
- Ethanol from lignocellulosic biomass is produced mainly via biochemical routes such as pretreatment, enzymatic hydrolysis, and fermentation

BIOMEDICAL WASTE

- Any waste which is generated during the **diagnosis, treatment or immunization of human beings or animals or in research activities** pertaining to that or in the production or testing of biological
- Biomedical waste, scattered in and around the hospitals invites flies, insects, rodents, cats and dogs that are responsible for the spread of communication diseases like plague and rabies
- Biomedical waste management Rules in 1998 – notified under EPA Act 1986

Types of Biomedical waste

1. **Human Anatomical Waste** ~ Human tissues, organs, body parts
2. **Animal Waste** ~ Animal tissues, organs, body parts, carcasses, bleeding parts, fluid, blood and experimental animals used in research, waste generated by veterinary hospitals and colleges, discharge from hospitals

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3. **Microbiology & Biotechnology Waste** ~ Wastes from laboratory cultures, stocks or specimen of live microorganisms, human and animal cell cultures used in research and infectious agents from research and industrial laboratories, wastes from production of biological, toxins and devices used for transfer of cultures
4. **Waste Sharps** ~ Needles, syringes, scalpels, blades, glass, etc. that may cause puncture and cuts. This includes both used and unused sharps

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5. **Discarded Medicine and Cytotoxic drugs** ~ Wastes comprising of outdated, contaminated and discarded medicines
6. **Soiled Waste** ~ Items contaminated with body fluids including cotton, dressings, soiled plaster casts, lines, bedding and other materials contaminated with blood
7. **Solid Waste** ~ Waste generated from disposable items other than the waste sharps such as tubing, catheters, intravenous sets, etc

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- 8. **Liquid Waste** ~ Waste generated from the laboratory and washing, cleaning, housekeeping and disinfecting activities
- 9. **Incineration Ash** ~ Ash from incineration of any biomedical waste
- 10. **Chemical Waste** ~ Chemicals used in production of biological, chemicals used in disinfecting, as insecticides, etc.

Management of Biomedical waste

- Incineration Technology: This is a high temperature thermal process employing combustion of the waste under controlled condition for converting them into inert material and gases
- Autoclaving: An autoclave is a machine that provides a physical method of sterilization by killing bacteria, viruses, and even spores present in the material. In general, an autoclave is run at a **temperature of 121° C** for at least 30 minutes
- Microwave irradiation: At a frequency of about 2450 MHz and a wavelength of 12.24 cm, microwaves destroy the majority of microorganisms
- Chemical Methods: 1% hypochlorite solution can be used for chemical disinfection

Colour coding of Biomedical waste

Colour	Waste	Treatment
Yellow	Human & Animal anatomical waste / Micro-biology waste and soiled cotton/dressings/linen/beddings etc. Incineration / Deep burial Red Tubing's, Catheters, IV sets.	Autoclaving / Microwaving/Chemical treatment
Red	Tubing's, Catheters, IV sets	Autoclaving / Microwaving/Chemical treatment
Blue/White	Waste sharps (Needles, Syringes, Scalpels, blades etc.	Autoclaving / Microwaving /Chemical treatment & Destruction / Shredding
Black	Discarded medicines/cytotoxic drugs,	Incineration ash, Chemical waste. Disposal in secured landfill



Proper segregation of wastes helps

- Minimizes the amount of potentially hazardous waste
- Facilitate proper packing and labeling of wastes
- Reduces occupational health and safety risks to the healthcare workers and rag pickers
- Improves infection control within the hospital
- Establish a uniform waste management practice

Self study

An overview of:

- CPHEEO guidelines for solid waste management
- Regulations for hazardous waste management in India